Synoptic Question 13: The Physics of an Electric Vehicle (EV)

Reading Passage

Read the following article and then answer the questions that follow.

Electric vehicles (EVs) represent a major shift in transportation technology, moving away from internal combustion engines to electric motors powered by large rechargeable batteries. The "fuel tank" of an EV is a high-voltage lithium-ion battery pack, typically rated in kilowatt-hours (kWh), which is a unit of energy.

This battery provides direct current (DC) electricity to an inverter, which converts it to alternating current (AC) to power one or more electric motors. These motors convert electrical energy into mechanical energy with high efficiency, providing instant torque for rapid acceleration.

One of the key features of an EV is regenerative braking. When the driver lifts off the accelerator or presses the brake pedal, the electric motor can operate in reverse as a generator. It converts the car's kinetic energy back into electrical energy, which is used to recharge the battery. This process recovers energy that would normally be lost as heat in conventional friction brakes. However, EVs also retain conventional hydraulic friction brakes (usually with brake discs) for rapid stops and to hold the car stationary.

Charging the battery is done by connecting the car to the electrical grid. This can be a slow "Level 2" AC charge at home or a very fast DC rapid charge at a public station, which can replenish the battery in under 30 minutes.

Data:

Data for a typical high-performance EV:

Table 1: Data for a Typical Electric Vehicle

Parameter	**Value**
Total mass of car and driver, m	1800 kg
Usable battery capacity, E_{batt}	85.0 kWh
Average battery voltage, V	400 V
Maximum range on full charge	450 km
Time for 0 to 100 km/h acceleration	4.5 s
Home (Level 2) charger power, P_{charge}	7.2 kW
Efficiency of home charging	90%
Efficiency of regenerative braking, η_{regen}	70%
Efficiency of electric motor, η_{motor}	95%
Mass of one front brake disc, m_{disc}	5.0 kg
Specific heat capacity of brake disc, c_{disc}	480 J kg^{-1} K^{-1}
Cost of electricity	\$0.22 per kWh

Questions

(a) The car accelerates from 0 to 100 km/h in 4.5 s.

(i) Convert 100 km/h to its SI unit (m s^{-1}).

(ii) Calculate the average acceleration of the car.

(iii) Calculate the average resultant force required to produce this acceleration.

(b) The battery stores energy in electrical form.

(i) Convert the 85.0 kWh usable battery capacity into joules. (1 kWh = 3.6×10^6 J).

(ii) Using the average voltage, calculate the total charge (in coulombs) that the usable part of the battery can deliver.

(c) The car has a maximum range of 450 km.

(i) Calculate the average energy consumption from the battery, in joules per metre (J m^{-1}).

(ii) Using the principle of work done, calculate the average total resistive force (air resistance, friction, etc.) acting on the car.

(d) The car is charged at home using a 7.2 kW charger.

(i) Calculate the time, in hours, required to charge the 85.0 kWh battery from 0% to 100%, assuming 100% efficiency.

(ii) Given the charging process is 90% efficient, calculate the actual time to fully charge the battery.

(iii) Calculate the total cost of this full charge.

(e) The car is braking, using its regenerative braking system.

(i) Calculate the loss in kinetic energy as the car slows from 80 km/h (22.22 m s^{-1}) to 30 km/h (8.33 m s^{-1}).

(ii) Calculate the amount of energy, in joules, that is recaptured and stored in the battery during this braking event. enumerate)

(f) During a sudden emergency stop from 80 km/h (22.22 m s^{-1}), the car uses only its conventional friction brakes.

(i) Assume 50% of the total kinetic energy is dissipated as thermal energy in the two front brake discs. Calculate the rise in temperature of each brake disc. enumerate)

(g) During a period of hard acceleration, the motor draws 400 A of current from the 400 V battery.

(i) Calculate the electrical power being supplied to the motor.

(ii) Calculate the mechanical power output of the motor.

Solutions

(a) (i) $v = 100 \text{ km/h} \times \frac{1000 \text{ m}}{1 \text{ km}} \times \frac{1 \text{ h}}{3600 \text{ s}} = 27.78 \text{ m s}^{-1}$.

(ii) $a = \frac{v-u}{t} = \frac{27.78-0}{4.5} = 6.17 \text{ m s}^{-2}$.

(iii) $F_{net} = ma = (1800)(6.17) = 11106 \text{ N} \approx 11100 \text{ N}$.

(b) (i) $E_{batt} = (85.0 \text{ kWh}) \times (3.6 \times 10^6 \text{ J/kWh}) = 3.06 \times 10^8 \text{ J}$.

(ii) From $E = QV$, the total charge is $Q = \frac{E_{batt}}{V} = \frac{3.06\times10^8}{400} = 765000 \text{ C}$.

(c) (i) Range $= 450 \text{ km} = 4.50 \times 10^5 \text{ m}$.
Energy per metre $= \frac{E_{batt}}{\text{range}} = \frac{3.06\times10^8}{4.50\times10^5} = 680 \text{ J m}^{-1}$.

(ii) Work done by motor = Resistive force × distance.
$E_{batt} = F_{resist} \times d$
$F_{resist} = \frac{E_{batt}}{d} = 680 \text{ N}$.

(d) (i) Time $= \frac{\text{Energy}}{\text{Power}} = \frac{85.0 \text{ kWh}}{7.2 \text{ kW}} = 11.81$ hours.

(ii) Actual Time $= \frac{\text{Energy needed by battery}}{\text{Power from wall}\times\eta} = \frac{85.0}{7.2\times0.90} = 13.1$ hours.

(iii) Energy drawn from grid $= \frac{\text{Energy stored}}{\eta} = \frac{85.0 \text{ kWh}}{0.90} = 94.44 \text{ kWh}$.
Cost $= (94.44 \text{ kWh}) \times (\$0.22/\text{kWh}) = \$20.78$.

(e) (i) $\Delta KE = \frac{1}{2}mv_i^2 - \frac{1}{2}mv_f^2 = \frac{1}{2}(1800)(22.22^2 - 8.33^2)$
$\Delta KE = 900(493.7 - 69.4) = 900(424.3) = 381870 \text{ J} \approx 3.82 \times 10^5 \text{ J}$.

(ii) $E_{recov} = \Delta KE \times \eta_{regen} = (381870) \times 0.70 = 267309 \text{ J} \approx 2.67 \times 10^5 \text{ J}$.

(f) (i) This is a stop from 80 km/h (22.22 m s^{-1}) to 0.
Total KE lost $= \frac{1}{2}mv_i^2 = \frac{1}{2}(1800)(22.22)^2 = 444330 \text{ J}$.
Energy dissipated in front brakes $Q_{front} = 444330 \times 0.50 = 222165 \text{ J}$.
Energy per disc $Q_{disc} = \frac{222165}{2} = 111083 \text{ J}$.
Using $Q = mc\Delta T$:
$\Delta T = \frac{Q_{disc}}{m_{disc}c_{disc}} = \frac{111083}{(5.0)(480)} = 46.28 \text{ K}$ (or 46.3 °C).

(g) (i) Electrical Power $P_{elec} = VI = (400)(400) = 160000 \text{ W} = 160 \text{ kW}$.

(ii) Mechanical Power $P_{mech} = P_{elec} \times \eta_{motor} = 160000 \times 0.95 = 152000 \text{ W} = 152 \text{ kW}$.

[Diagram showing EV energy flow: (1) Acceleration: Battery to Motor to Wheels. (2) Regen Braking: Wheels to Motor (Generator) to Battery.]

Figure 1: Energy Flow in an Electric Vehicle

This document keeps the uploaded file `uploads/gemini_EV.pdf` as the base source and appends a new **Variant 2** set in CIE 9702 Paper 4 style, with higher-difficulty and open-ended physics tasks.

Reading Passage (Variant 2)

Read the passage and extract relevant numerical information for the questions.

The test vehicle is a medium-mass electric car with driver, with a total mass of 1850 kg. It stores 82.0 kWh of usable electrical energy in a battery pack operating at a nominal 380 V. Under steady cruising conditions, the reported maximum range is about 420 km.

The body is designed for low drag, with drag coefficient $C_d = 0.24$ and frontal area $2.35\,\mathrm{m}^2$. For this paper, take air density as $1.20\,\mathrm{kg\,m^{-3}}$ and rolling-resistance coefficient as 0.010. Assume gravitational field strength $9.81\,\mathrm{m\,s^{-2}}$. The drivetrain from battery to wheels operates at 90% efficiency, while regenerative braking returns 68% of available braking energy to the battery.

For charging analysis, a DC charger rated at 120 kW is used. Grid-to-battery charging efficiency is 92%, and electrical energy is billed at \$0.24 per kWh.

Data for Variant 2

Use the following data unless otherwise stated.

Parameter	Value
Total mass of car + driver, m	1850 kg
Usable battery capacity, E_{batt}	82.0 kWh
Nominal battery voltage, V	380 V
Range at steady cruising speed	420 km
Drag coefficient, C_d	0.24
Frontal area, A	$2.35\,\mathrm{m}^2$
Air density, ρ	$1.20\,\mathrm{kg\,m^{-3}}$
Rolling-resistance coefficient, C_{rr}	0.010
Gravitational field strength, g	$9.81\,\mathrm{m\,s^{-2}}$
Drivetrain efficiency (battery to wheels), η_d	0.90
Regenerative-braking efficiency, η_r	0.68
Maximum charging power (DC), P_{ch}	120 kW
Grid-to-battery charging efficiency	92%
Electricity tariff	\$0.24 per kWh

Question 13 (Variant 2)

An electric vehicle (EV) travels on a motorway and through hilly terrain. Assume still air unless stated.

(a) The EV cruises steadily at $30.0\,\mathrm{m\,s^{-1}}$ on level road.

(i) Show that the aerodynamic drag force is approximately 3.05×10^2 N.

(ii) Calculate the rolling-resistance force.

(iii) Hence determine the mechanical power required at the wheels.

(iv) Determine the electrical power drawn from the battery.

(b) The EV then climbs a slope of angle θ, where $\sin\theta = 0.060$, while maintaining $22.0\,\mathrm{m\,s^{-1}}$.

(i) Calculate the additional force needed to overcome component of weight down the slope.

(ii) Calculate the total wheel power needed during the climb.

(iii) Explain briefly why energy consumption per kilometre is much higher on this section.

(c) During descent, the EV slows from $28.0\,\mathrm{m\,s^{-1}}$ to $16.0\,\mathrm{m\,s^{-1}}$ while dropping vertical height $75\,\mathrm{m}$.

(i) Calculate the change in kinetic energy of the vehicle.

(ii) Calculate the loss in gravitational potential energy.

(iii) Hence determine the maximum energy available to braking systems.

(iv) Estimate the energy returned to the battery by regeneration.

(d) Open-ended design task (conceptual + quantitative):

The manufacturer wants to increase real-world range by at least 12% without increasing battery capacity.

Propose and evaluate **two distinct physics-based strategies**. Your answer should:

(i) identify the physical principle for each strategy,

(ii) include one quantitative estimate for each strategy,

(iii) discuss one practical trade-off for each strategy,

(iv) conclude which strategy is likely to be more effective in mixed driving.

Sample Solution (Variant 2)

(a)

$$\begin{aligned} F_d &= \tfrac{1}{2}\rho C_d A v^2 = \tfrac{1}{2}(1.20)(0.24)(2.35)(30.0)^2 \\ &= 3.05 \times 10^2\,\mathrm{N}\ (3\ \text{s.f.}) \end{aligned}$$

$$F_{rr} = C_{rr} mg = (0.010)(1850)(9.81) = 1.81 \times 10^2\,\mathrm{N}$$

Total resistive force on level road:

$$F_{\text{tot}} = F_d + F_{rr} = 305 + 181 = 486\,\mathrm{N}$$

Mechanical wheel power:

$$P_{\text{wheel}} = F_{\text{tot}} v = (486)(30.0) = 1.46 \times 10^4\,\mathrm{W} = 14.6\,\mathrm{kW}$$

Electrical battery power:

$$P_{\text{batt}} = \frac{P_{\text{wheel}}}{\eta_d} = \frac{14.6}{0.90} = 16.2\,\mathrm{kW}$$

(b)

$$F_g = mg\sin\theta = (1850)(9.81)(0.060) = 1.09 \times 10^3\,\mathrm{N}$$

At $22.0\,\mathrm{m\,s^{-1}}$, aerodynamic drag becomes

$$F_d = \tfrac{1}{2}(1.20)(0.24)(2.35)(22.0)^2 = 1.64 \times 10^2\,\mathrm{N}$$

Rolling resistance approximately unchanged:

$$F_{rr} = 1.81 \times 10^2\,\mathrm{N}$$

Total opposing force up slope:

$$F_{\mathrm{tot,climb}} = F_g + F_d + F_{rr} = 1089 + 164 + 181 = 1.43 \times 10^3\,\mathrm{N}$$

Wheel power on climb:

$$P_{\mathrm{wheel,climb}} = F_{\mathrm{tot,climb}} v = (1434)(22.0) = 3.15 \times 10^4\,\mathrm{W} = 31.5\,\mathrm{kW}$$

Battery power on climb:

$$P_{\mathrm{batt,climb}} = \frac{31.5}{0.90} = 35.0\,\mathrm{kW}$$

Explanation: additional continuous work is done against gravity, so energy per unit distance increases by approximately $mg\sin\theta$ per metre, making hill climbing far more energy intensive than level cruising.

(c)

$$\Delta E_k = \tfrac{1}{2}m(v_f^2 - v_i^2) = \tfrac{1}{2}(1850)(16.0^2 - 28.0^2) = -4.88 \times 10^5\,\mathrm{J}$$

So kinetic energy decrease magnitude is 4.88×10^5 J.

$$\Delta E_p = mg\Delta h = (1850)(9.81)(75) = 1.36 \times 10^6\,\mathrm{J}$$

Maximum energy removed by braking systems (neglecting drag/rolling during interval):

$$E_{\mathrm{brake,max}} = |\Delta E_k| + \Delta E_p = 4.88 \times 10^5 + 1.36 \times 10^6 = 1.85 \times 10^6\,\mathrm{J}$$

Energy stored back in battery by regeneration:

$$E_{\mathrm{return}} = \eta_r E_{\mathrm{brake,max}} = (0.68)(1.85 \times 10^6) = 1.26 \times 10^6\,\mathrm{J}$$

(d) Open-ended sample response

Strategy 1: Reduce aerodynamic drag (C_dA).

Physics principle: drag force varies as $F_d \propto C_dAv^2$, and drag power as $P_d \propto C_dAv^3$.

Estimate: if redesign lowers effective C_dA by 15%, then cruising drag power drops by 15%. At $30\,\mathrm{m\,s^{-1}}$, drag was about 9.15 kW (from F_dv), so saving is about 1.37 kW. Against baseline battery draw 16.2 kW, this is about 8.5% reduction, implying roughly 9% range gain in high-speed driving.

Trade-off: tapered rear and active shutters may increase manufacturing complexity/cost and can reduce cabin or luggage volume.

Strategy 2: Lower rolling resistance and mass.

Physics principle: rolling loss $F_{rr} = C_{rr}mg$ and inertial losses scale with mass in stop-start motion.

Estimate: reducing C_{rr} from 0.010 to 0.008 lowers rolling force by 20%. Baseline $F_{rr} = 181$ N, so saving is 36 N. At $30\,\mathrm{m\,s^{-1}}$, wheel power saving is 1.08 kW; battery saving about 1.20 kW (7.4 % of baseline battery power). A further 8% vehicle mass reduction improves hill-climb and transient energy demands.

Trade-off: low rolling-resistance tyres may reduce peak grip; lightweight materials can raise cost and repair complexity.

Conclusion: for mixed driving, combined drag reduction + moderate rolling/mass improvements can exceed 12%. If one strategy must be prioritised, drag reduction gives larger benefit at motorway speeds, while rolling/mass measures offer broader benefits in urban and hilly routes.

Mark Scheme (Variant 2 only)

CIE 9702-style annotation used: M = method mark, A = accuracy mark, B = independent point mark, E = evaluation/quality of explanation.

(a) [8]

1. Uses $F_d = \frac{1}{2}\rho C_d A v^2$ correctly. M1
2. $F_d \approx 3.05 \times 10^2\,\mathrm{N}$ (accept 300–310 N). A1
3. Uses $F_{rr} = C_{rr} mg$. M1
4. $F_{rr} \approx 1.81 \times 10^2\,\mathrm{N}$. A1
5. Adds resistive forces correctly. M1
6. $P_{\text{wheel}} = Fv$ with correct substitution. M1
7. $P_{\text{wheel}} \approx 14.6\,\mathrm{kW}$. A1
8. Uses efficiency relation for battery power and obtains $16.2\,\mathrm{kW}$ (± 0.3 kW). A1

(b) [7]

1. Uses $F_g = mg\sin\theta$. M1
2. $F_g \approx 1.09 \times 10^3\,\mathrm{N}$. A1
3. Correctly recalculates drag at $22\,\mathrm{m\,s^{-1}}$. M1
4. $F_d \approx 1.64 \times 10^2\,\mathrm{N}$ (accept 160–170 N). A1
5. Determines total opposing force including rolling term. M1
6. Correct climb wheel power ($31.5\,\mathrm{kW}$ approx). A1
7. Valid physics explanation of increased energy per distance due to continuous gain in GPE/work against weight component. B1

(c) [6]

1. Correct kinetic-energy change expression and sign handling. M1
2. Magnitude $|\Delta E_k| \approx 4.88 \times 10^5\,\mathrm{J}$. A1
3. Correct gravitational potential-energy change expression. M1
4. $\Delta E_p \approx 1.36 \times 10^6\,\mathrm{J}$. A1
5. Combines energies consistently to find braking-system energy. M1
6. Regenerated energy $\approx 1.26 \times 10^6\,\mathrm{J}$ using 68%. A1

(d) Open-ended [9]

Award up to 9 marks using levelled points below.

1. Two distinct valid strategies identified (e.g. aerodynamics, tyres/rolling resistance, mass reduction, thermal management, control strategy). B2
2. Correct physics principles for both strategies. B2
3. One quantitative estimate for each strategy with sensible magnitude and correct physics links. M2
4. Trade-off/limitation discussed for each strategy. E2
5. Reasoned final judgement for mixed driving, supported by earlier analysis. E1

Total: 30 marks